University of Liege

2023-2024

High-Dimensional Statistics 2023-2024

Project: Dependence structure and linear modeling

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December 11, 2023

1 Graphical model of the dependence structure between the score variables

1. Assuming that the multivariate normality assumption holds for the data base restricted to the scores of each test separately (the 5 scores of test Saber11 on one side and the 6 scores of test SaberPro on the other side), the plots of the number of edges that would be included in each graph with respect to the value of the L1-regularization parameter are represented in the figure 1.

2. Fixing to 5 the number of edges to represent in each graph, the both graphical models are represented are represented in the figure 2.

   Given the previous plot relative to Saber_S11 test, all the values in the range [71;75] allow to obtain only 5 edges for the graphical model of Saber_S11 test. Thereby one may choose 75 as regularization parameter value for getting 5 edges. While whether the regularization value is set in the range [407;410], we get also 5 edges for the one of Saber_PRO test. Thus we can pick 410 as regularization parameter value in order to satisfy the question requirements.

3. Also, due to the multivariate normality assumption, we can infer some conclusions if we take into account five edges in each graphical model:

   Differences:

   - In the graphical model relative to the Saber_S11 test, none of the vertices is isolated. That is not the case in the one relative to the Saber_PRO test where one can see how the WC_PRO vertex is isolated, meaning that the WC_PRO feature is conditionally independent of the other features linked to the same Saber_PRO test. Moreover, this implies weak correlation values among her and the other features, rendering PCA less efficient for projecting the data in 2-D. In the first project, it was difficult to attain 70% of cumulative variance explained by the two first PCs.
   - In the Saber_PRO test graphical model, the CC_PRO and CR_PRO vertices are not conditionally independent, meaning that these both courses are dependent (related courses). But in the one relative to the Saber_S11, CC_S11 and CR_S11 are conditionally independent, meaning the both courses are not related in high secondary school, even if CC_PRO and CR_PRO are their sequel at university.
   - Furthermore, all the vertices are dependent on the ENG_S11 vertex in the Saber_S11 test graphical model. This is not the case for the Saber_PRO test one.

   Similarities: In both graphical models, there is always a vertex triplet that is dependent (CC_PRO, CR_PRO, and FEP_PRO vertices in the Saber_PRO test graphical model, and BIO_S11, ENG_S11, and MAT_S11 in the other).

   Besides, we notice that MAT_S11, CR_S11, and CC_S11 score-variable kurtosis values are larger than 3, while the one of ENG_S11 and BIO_S11 are a little bit close to 3. This can lead to think the former score-variables do not follow a normal distribution, and the latter seem to follow the normal distribution. However, none of their skewness values is close to zero. Meaning, even as there is a little bit deviation of these previous

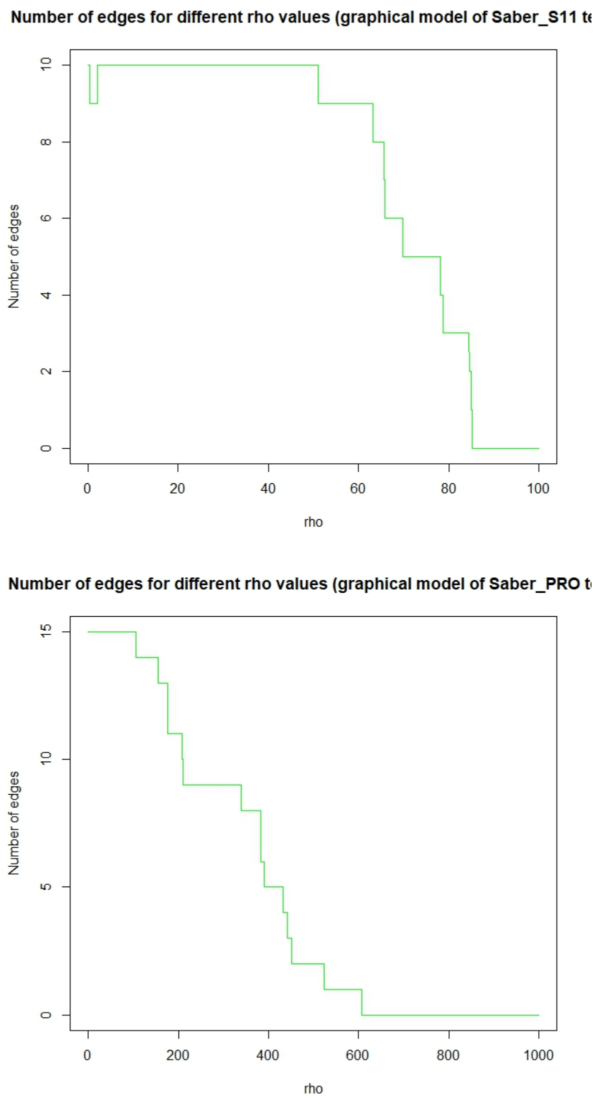

Figure 1: Plots of the number of edges according to the regularization parameter (left the one of Saber_S11 test and right the one of Saber_PRO test.

Figure 2: Left graphical model of Saber_S11 test and right the one of Saber_PRO test.

values with the kurtosis and skewness values of a normality assumption, a multivariate normality assumption seems to hold for the Saber_S11 test variables. One can rely over the scatter matrix diagonal to observe the kernel density of the distribution of each variable (figure 3) .

However, the Saber_PRO test score-variable kurtosis and skewness values do not match the values of a normality assumption mentioned earlier. If we assume the vector consists of all these score-variables is a Gaussian vector, then all linear combinations of its score-variables would follow a normal distribution; meaning that each score-variable would follow a normal distribution. But it is not the case at hand with the Saber_PRO test score-variables. Thus, the multivariate normality assumption does not hold and we group some kurtosis and skewness values of some score-variables in the table 1.

	MAT_S11	QR_PRO	CR_PRO	ENG_S11	BIO_S11
Skewness	0.4896417	-0.8454640	-0.3238777	0.7969504	0.2749560
Kurtosis	2.673837	3.270783	1.814454	3.025270	2.818867

Table 1: Skewness and Kurtosis values of some score-variables.

2 Linear modeling of the SaberPro scores

1. As we saw it in the first project, the overall Saber_S11 test score-variables are not strongly correlated (figure 3). The potential issues of multicollinearity among these score-variables are the following:

    - Estimated regression coefficient instability: In fact, a small variation in the data might lead to changing significantly the estimated coefficients.
    - Difficulty to isolate the effect of each variable: Indeed, when two or more independent variables are strongly correlated, it becomes difficult to determine the effect

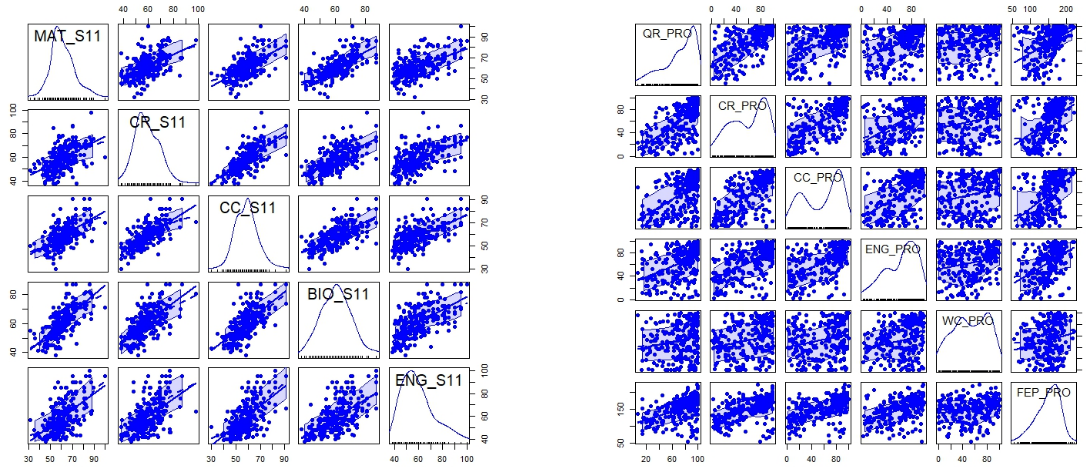

Figure 3: Left scatter matrix of Saber_S11 variables test and right the one of Saber_PRO test variables

of each of them on the dependent variable.

2. In the first project, you have deleted outlying observations because they lead to a corrupted empirical covariance matrix and subsequently to a wrong analysis. We start by dropping them also in this project with MCD technique while starting by standardizing the data.

   According to the R-squared values in the table 2 below, the most promising results are R-squared values obtained from the linear regression of ENG_PRO and CR_PRO score-variables, while the one with the least promising R-squared value is obtained from WC_PRO score-variable. Therefore, we keep WC_PRO, ENG_PRO, and CR_PRO variables for the next of our analysis.

Models	`mod_QR_PRO`	`mod_CR_PRO`	`mod_ENG_PRO`	`mod_CC_PRO`	`mod_FEP_PRO`	`mod_WC_PRO`
R^2-measure	0.455	**0.472**	**0.560**	0.422	0.255	**0.073**

Table 2: R^2-measure Values for Different Models Relative to the SaberPro Scores

3. **Selection of Variables According to AIC (Backward Stepwise)** The goal is to minimize the AIC, and thus the model that supplies the smallest AIC value is considered the optimal model.

   CC_S11 and ENG_S11 are score-variables that are selected as predictors for the linear modeling of WC_PRO score-variable. We have the same score-variables selected as in the previous selection form the one of ENG_PRO score-variable. BIO_S11, CC_S11, and CR_S11 are score-variables chosen for the one of CR_PRO score-variable.

   Hence we obtain the following simplified models : The means of the variables are given by the following linear models:

$$\text{Mean of WC\_PRO} = 10.296 + 0.393 \cdot \text{CC\_S11} + 0.312 \cdot \text{ENG\_S11}$$

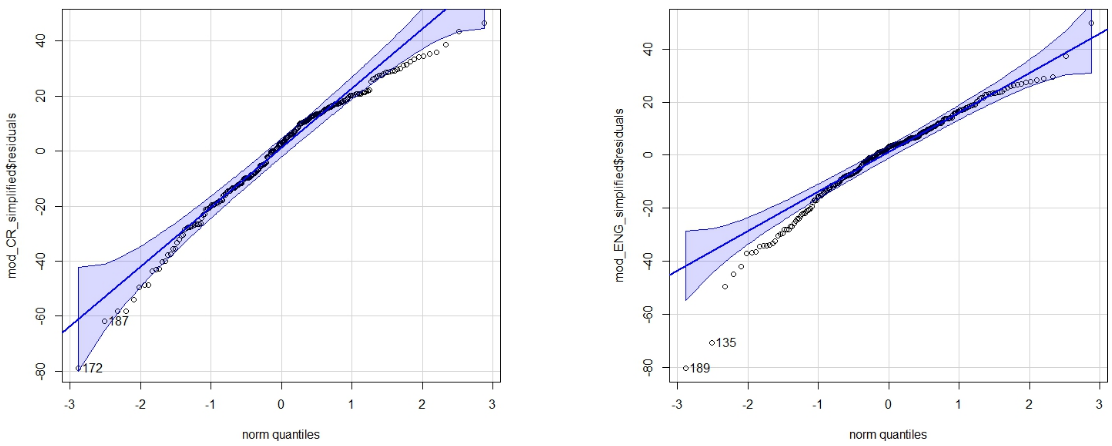

Figure 4: Checking normality assumptions of the residuals gotten from the linear regression models with CR_PRO (left) and ENG_PRO (right) variables as dependent variables.

$$\text{Mean of ENG\_PRO} = -35.757 + 0.478 \cdot \text{CC\_S11} + 1.179 \cdot \text{ENG\_S11}$$

$$\text{Mean of CR\_PRO} = -73.836 + 0.869 \cdot \text{CC\_S11} + 0.434 \cdot \text{CR\_S11} + 0.927 \cdot \text{BIO\_S11}$$

while assuming their associated residuals follow the normal distribution with 0 mean and constant variance σ^2.

4. Checking out the hypotheses required for the residuals derived from the simplified models : It turns out that the constrained models obtained are better than theîr corresponding full models in our analysis through ANOVA test. Morever, in the qq-plots below, the points do not lie on the straight line, and also, according to the p-value of its Shapiro-Wilk test, which is strictly smaller than 5%, the normality of the residuals is rejected. Moreover, the variation inflation factor value of each explanatory variable of these simplified models is strictly smaller than 10. Thus, there is no issue of multicollinearity among its explanatory variables.

   Form left to right, the first histogram is skew while the next resembles to the mixture of two Gaussian distributions (figure 5).

5. To conclude, we have observed that after performing some dependent variable selection among the test Saber_PRO score-variables through the R-squared given by the full linear regression models, and after optimizing each linear model via backward stepwise selection aiming to select the model with the smallest AIC yielding the optimal model, their residual normality assumptions are not satisfied. This means the results obtained via these linear regressions do not hold.

   Thus, the interpretation that we infer from these outcomes is that university programs tend to develop other skills, implying that scores derived after such training might not

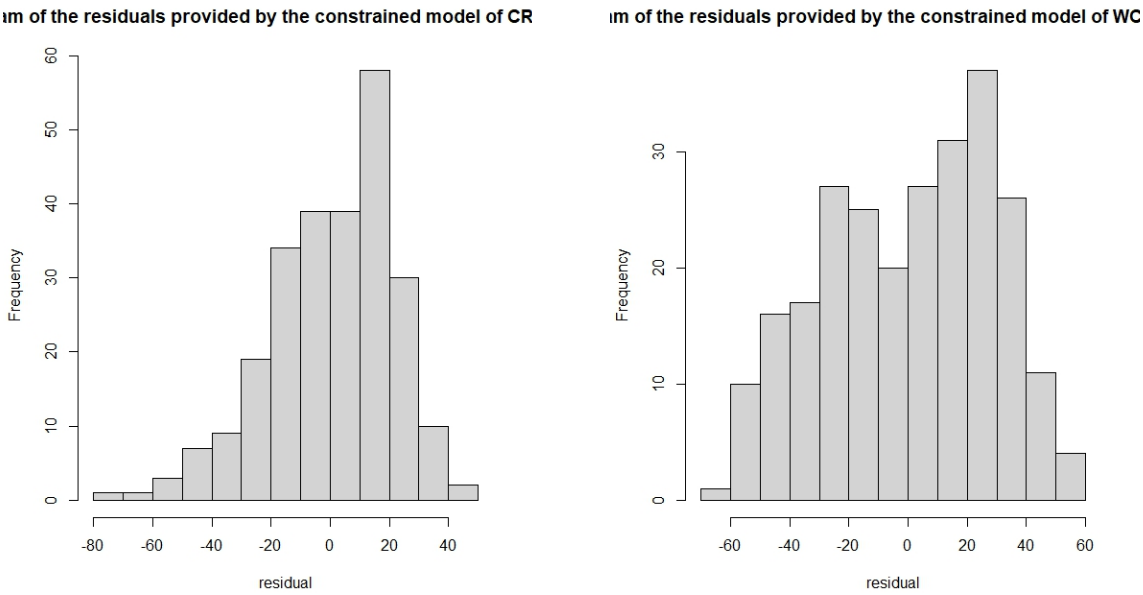

Figure 5: Histograms of the residuals gotten from linear regression models with CR_PRO (left) and WC_PRO (right) variables.

be influenced by previous results, and we can turn towards generalized linear regression for modeling the Saber_PRO scores.